Mode-Selection / Cross-Validation

MADRE: Méthodologie pour l'Analyse de Données Réseaux
(Methodology for Network Data Analysis)

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Outline

1. Linear and polynomial regression
2. Prediction errors and accurancy
3. Variance-Bias trade off
4. Resampling Methods

Test your knowledge on linear regression

1. Allez sur wooclap.com
2. Entrez le code d'événement dans le bandeau supérieur

Code d'événement
MADRE3

Polynomial regression

The linear regression model assumes a linear relationship between the response and the input (predictors).
But! the true relationship may be non-linear.
We can extend the linear model, using polynomial regression.
For 1-D input x we write an l-polynomial model:

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + ... + \beta_l x^l + \epsilon$$

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$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + ... + \beta_l x^l + \epsilon$$

But it is still a linear model for the parameters!
It is just a multiple linear regression $(x_1 := x, x_2 := x^2, ..., x_l := x^l)$
Use standard linear regression software!

Polynomial regression

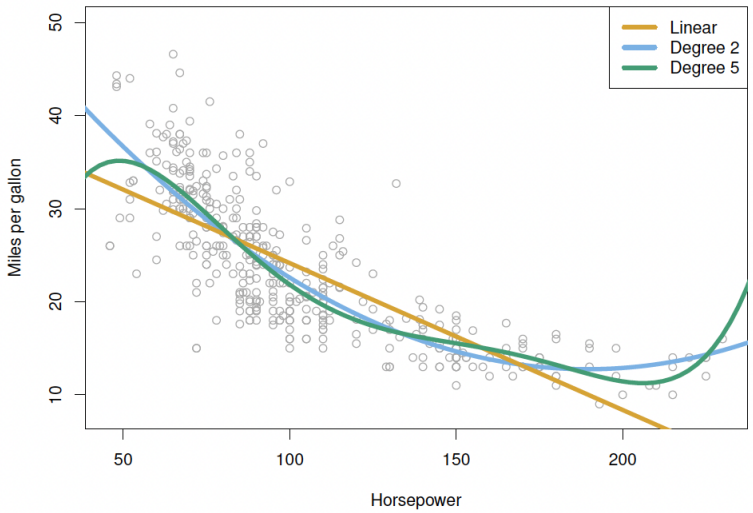

Figure: Polynomial vs Linear fit[1].

[1] Gareth James, Daniela Witten, Trevor Hastie, Robert Tibshirani. "An introduction to statistical learning: with applications in R". Springer Texts in Statistics.

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Prediction Errors

Linear predictions cannot be accurate but will always have an irreducible error, no matter how good the choice of the predictor $\hat{f}$

$$\mathbb{E}[(y-\hat{y})^2] = \mathbb{E}[(f(x)-\hat{f}(x))^2] + \mathbb{E}[\epsilon^2] +$$

$$+2\mathbb{E}[\epsilon(f(x)-\hat{f}(x))] =$$

$$\mathbb{E}[(f(x)-\hat{f}(x))^2] + Var(\epsilon)$$

We have 2 types of errors...

Reducible vs Irreducible

- The irreducible error, due to the random error ϵ in the model, whose variance is unknown.
- The reducible error, due to errors in the estimate of the model parameters $\hat{\beta}_i$.This type of error can be reduced by using :
  - larger sample sets when estimating the coefficients
  - different models $\hat{f}(x)$ that better describe the unknown function f (x) (could be non-linear).

In practice larger intervals are used for prediction to account for both types of errors.

Accurancy

The MSE is a measure of model accuracy. For the available data set $D_n = ((x_1, y_1), (x_2, y_2), ..., (x_n, y_n))$,

$$MSE(D_n; \hat{\beta}_1, \hat{\beta}_0) = \frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{\beta}_1 x_i - \hat{\beta}_0)^2$$

We guarantee maximum accuracy by

$$\min_{\hat{\beta}_1, \hat{\beta}_0} MSE(D_n; \hat{\beta}_1, \hat{\beta}_0)$$

By choosing $\hat{\beta}_1$ and $\hat{\beta}_0$ that minimize MSE we reduce the reducible part, but cannot change the irreducible part due to noise.

Train MSE

The predicted response $\hat{y}(x_i) = \hat{\beta}_1 x_i + \hat{\beta}_0$ will be close to y_i , because the parameters are chosen to minimise their difference! We say that the model is trained with data D_n.

$$MSE_{train} = MSE(D_n; \hat{\beta}_0, \hat{\beta}_1)$$

But! We actually want that the model predicts good values for unknown data, $x_o \notin D_n$.

Test MSE

We need a different test data set D_m^{test} with a number $m > 1$ of samples, to test the accuracy of our prediction model. For this test data set, we relate the accuracy metric

$$MSE_{test} := MSE(D_m^{test}; \hat{\beta}_0, \hat{\beta}_1) \neq MSE_{train}$$

Question 1: How good does our "minimum MSE linear predictor" behave for the test data set?
Question 2: If we use other prediction models, e.g. non-linear, can these predict better for the same test data set?

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Flexibility VS Interpretability

Question 2: If we use other prediction models, e.g. non-linear, can these predict better for the same test data set?

Higher order polynomial models offer more flexibility.

- In the most extreme case we can use a model whose curve passes through every point of the train data set D_n. We can propose a polynomial fit with $l = card(D_n)$.
  Is this a good predictive model?
- At the other extreme we can use a very restrictive model (simple linear regression), with $l = 1$.
  Maybe this simple model is better?

Example 1

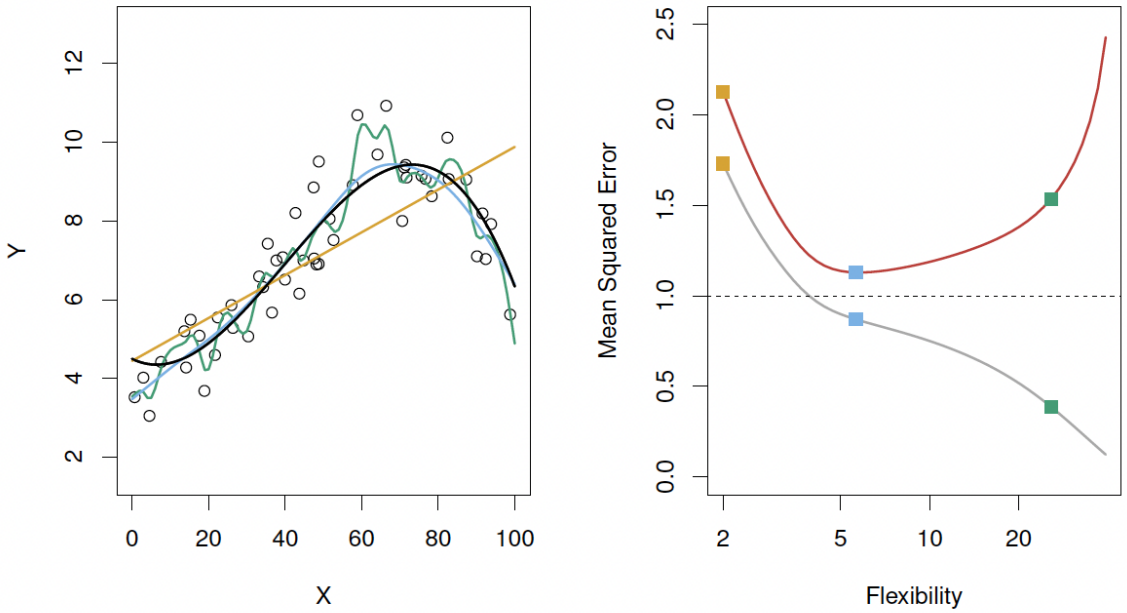

Figure: Black curve is the real one[2]

[2] Gareth James, Daniela Witten, Trevor Hastie, Robert Tibshirani. "An introduction to statistical learning: with applications in R". Springer Texts in Statistics.

Example 2

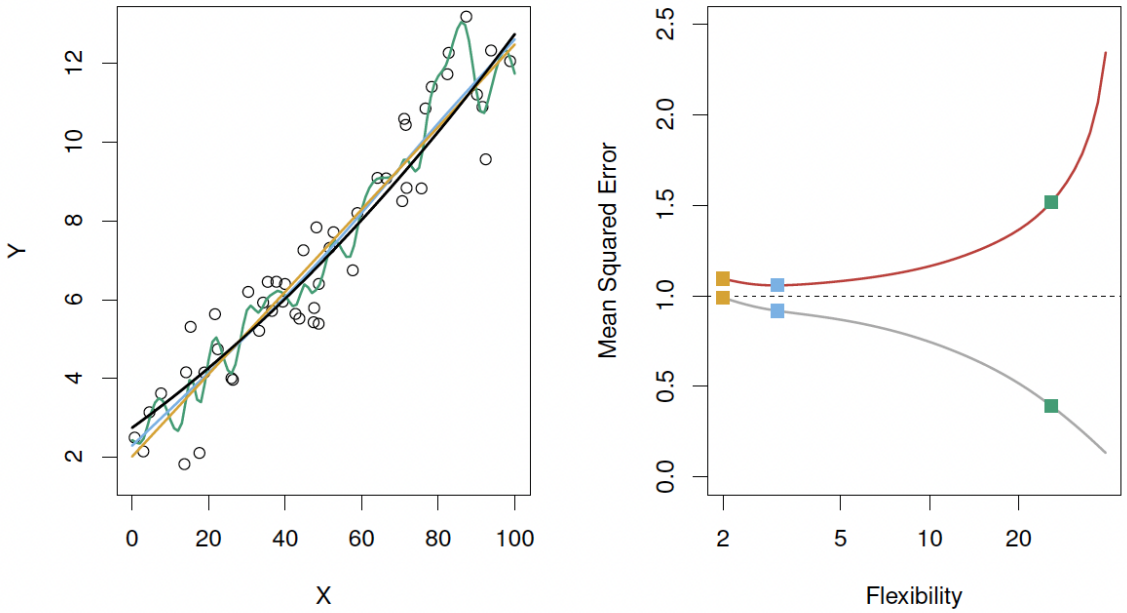

Figure: Black curve is the real one [3]

[3] Gareth James, Daniela Witten, Trevor Hastie, Robert Tibshirani. "An introduction to statistical learning: with applications in R". Springer Texts in Statistics.

Example 3

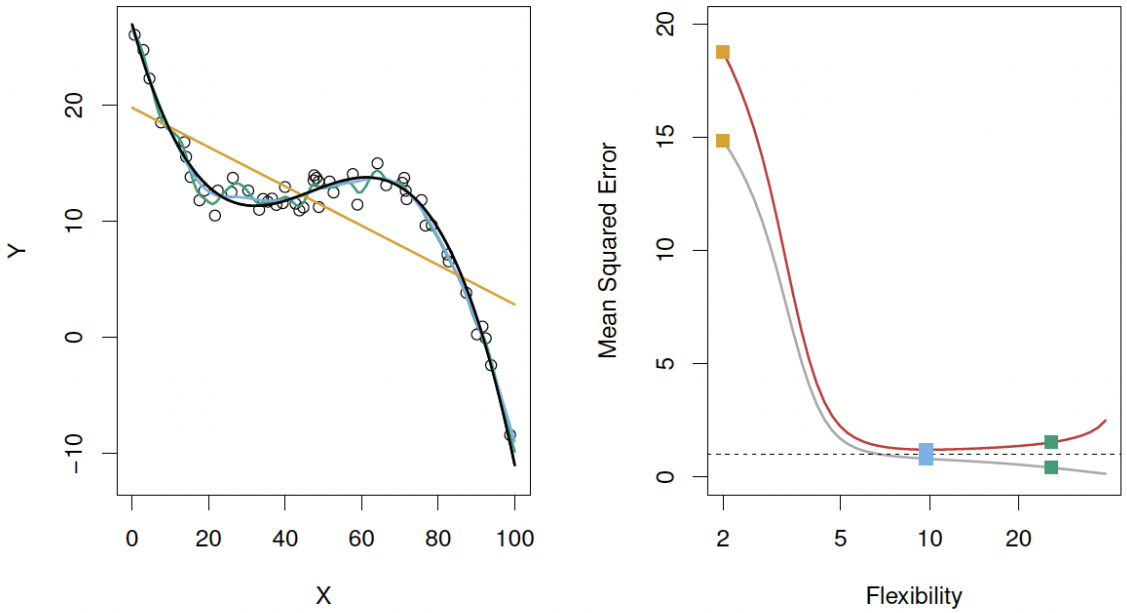

Figure: Black curve is the real one [4]

[4] Gareth James, Daniela Witten, Trevor Hastie, Robert Tibshirani. "An introduction to statistical learning: with applications in R". Springer Texts in Statistics.

Optimal model selection

Question 1: How good does our "minimum MSE linear predictor" behave for the test data set?

The MSE_{test} is always higher than noise (irreducible error).

Optimal model selection

Question 2: If we use other prediction models, e.g. non-linear, can these predict better for the same test data set?

The more flexibility, the lower the MSE_{train}.
But! The MSE_{test} always has a **U** shape (fundamental property) with respect to degree (x-axis). The optimal mode is the one that minimizes the MSE_{test} : trade-off between flexibility vs interpretability.
We call this the **Variance VS Bias trade-off**.

Overfitting/Underfitting

Overfitting: Small MSE_{train} but large MSE_{test} . The statistical learning model picks patterns that are caused by randomness rather than the true properties of the unknown $f(x)$.
needs lower flexibility!

Underfitting: Large MSE_{train} and large MSE_{test}. The learning model is too rigid to accurately describe the unknown $f(x)$.
needs higher flexibility!

Numerical Example

Consider the numerical example with
$D_n = \{(1,3),(2,4),(3,8),(4,14)\}$

- With the first three set elements do linear regression.
- Use the forth element to derive the $MSE_{test-linear}$.
- With the first three set elements do quadratic regression.
- Use the forth element to derive the $MSE_{test-quadratic}$.

Which method is best?

Numerical Example

We use the set $D_3 = \{(1,3),(2,4),(3,8)\}$ for the linear regression, and we get:

$$\hat{y} = 2.5x + 0$$

The MSE is calculated using the test set $\{(4,14)\}$

$$MSE_{test-linear} = (14 - 2.5 * 4 - 0)^2 = 16$$

For the quadratic regression we get:

$$\hat{y} = 1.5x^2 - 3.5x + 5$$

The MSE calculated using the test is:

$$MSE_{test-quadratic} = (14 - 1.5 * 16 + 3.5 * 4 - 5)^2 = 1$$

$MSE_{test-quadratic} < MSE_{test-linear}$ so the quadratic fit is better!

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Resampling methods

In all the above, we assumed that two disjoint data-sets are available:

- a train data-set D_n,
- a test data-set D_m^{test}

where $Dn \cap D_{test} = \emptyset$

However, usually we only have one data-set available D_n, to both train and test the machine learning algorithm.

What we should do?
Use resampling methods as **Cross-validation** and **Bootstrapping**.

Cross-validation

In includes:

- The Validation Set Approach
- Leave-One-Out Cross-Validation (LOOCV)
- k-Fold Cross-Validation

The Validation Set Approach

The Validation Set Approach is splitting your train dataset into parts: training and validation set.
The data subset are:

- **training set**: The sample of data used to fit the model. (e.g. weights and biases in the linear model or neural network)
- **validation set**: The sample of data used to provide an unbiased evaluation of a model fit on the training dataset while fine-tuning model hyper-parameters. (frequent evaluations)
- **test set**: The sample of data used to provide an unbiased evaluation of a final model fit on the training dataset. (only used once a model is completely trained)

Sometimes we can use the validation set to test our model (as a test set) but is not good practice.

Data split

Consider that a test set is provided independently of observed data.

The approach

It's a Naive approach.
Split the observation data set Dn in two:

- a train set
- a validation set

e.g. 80% of the elements of D_n belong to the train and the rest 20% to the validation set.

- Use the train set to fit the model.
- Use the validation set to evaluate performance, e.g. MSE_{test}.

Drawbacks

The method has two main problems:

- The test error rate depends on the data split $\rightarrow MSE_{test}$ can be highly variable!
- Not all available n data are used for training $\rightarrow$ worse performance with less observations, and MSE_{test} is overestimated!

Validation set example

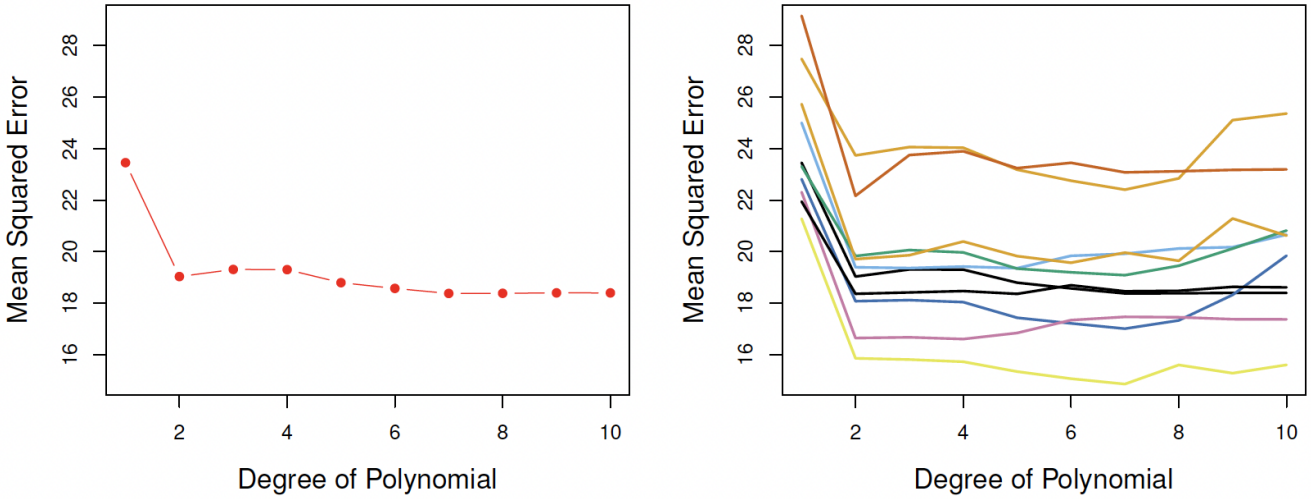


Figure: Variability of the MSE_{test} depending on the data split[5]

[5] Gareth James, Daniela Witten, Trevor Hastie, Robert Tibshirani. "An introduction to statistical learning: with applications in R". Springer Texts in Statistics.

Leave-One-Out Cross-Validation (LOOCV)

Split the data set D_n again in two:

- a validation set of a single observation (x_1, y_1)
- a train set of the rest $n-1$ observations.

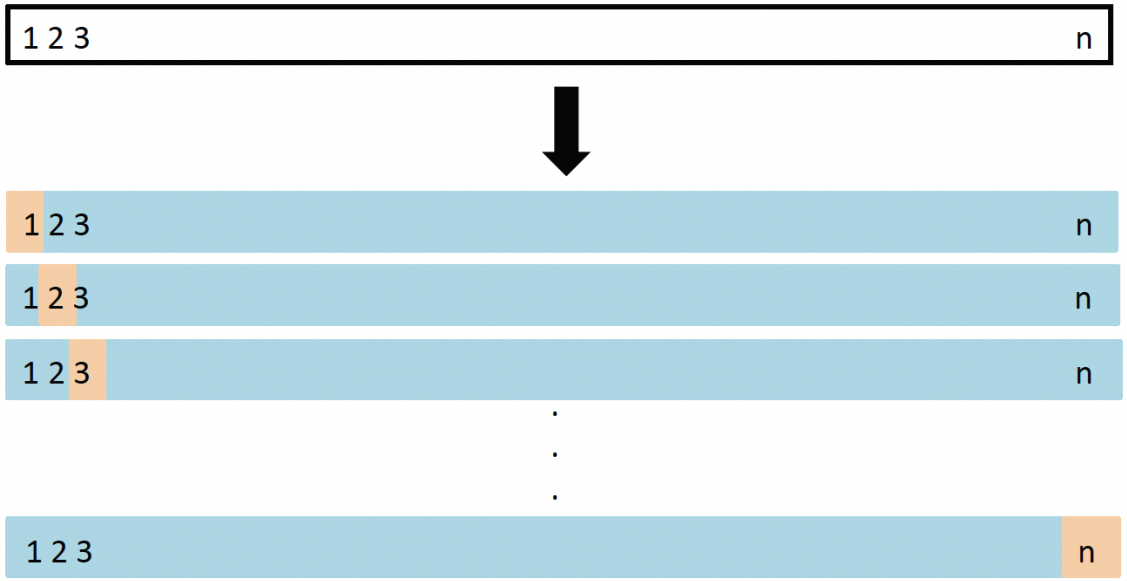
A prediction $\hat{y}_1$ is made only for the excluded observation using x_1.

$$MSE_{test} = MSE_1 = (y_1 - \hat{y}_1)^2.$$

Problem: The evaluation is based on a single observation
$\rightarrow$ highly variable.

LOOCV x n

Solution: repeat n times fot n different splits[6].



[6] Gareth James, Daniela Witten, Trevor Hastie, Robert Tibshirani. "An introduction to statistical learning: with applications in R". Springer Texts in Statistics.

LOOCV x n

1	2 3 ... n-1 n	MSE_1
2	1 3 ... n-1 n	MSE_2
3	1 2 ... n-1 n	MSE_3
...	...	...
n	1 2 3 ... n-1	MSE_n

The LOOCV estimate for the test MSE is the average of these n test error-estimates

$$CV_{(n)} = \frac{1}{n}\sum_{i=1}^{n} MSE_i.$$

No randomness in the result + uses all observations!

k-Fold Cross-Validation

For large n LOOCV is time-consuming to calculate all MSE_i . In this case is better to use k-fold Cross-Validation:

- The data-set Dn is split into $1 \leq k \leq n$ folds.
- The 1st fold is treated as validation set and the rest $k-1$ for training $\rightarrow MSE_1$ is caclulated.
- Repeat k times by choosing a different fold for validation set each time $\rightarrow MSE_k$ is calculated.

The k-fold estimate is computed by averaging:

$$CV_{(k)} = \frac{1}{n}\sum_{i=1}^{k} MSE_i.$$

k-Fold x *k*

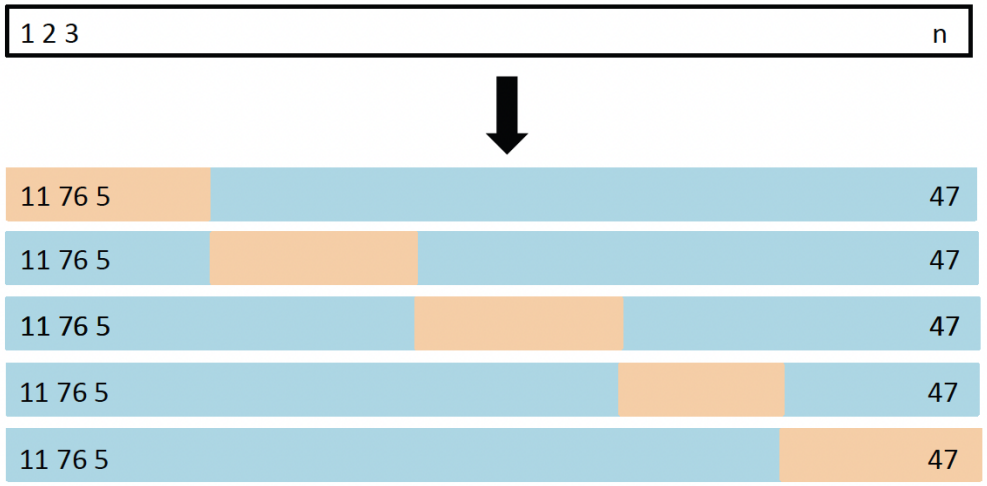


Figure: Repeat k times, for k different splits5[7].

[7] Gareth James, Daniela Witten, Trevor Hastie, Robert Tibshirani. "An introduction to statistical learning: with applications in R". Springer Texts in Statistics.

Computational advantages

- LOOCV is a special case of k-fold CV, for $k = n$.
- In practice $k = 5$ or $k = 10$.
- k-fold CV with $k < n$ can have lower variance in the MSE, than LOOCV.

LOOCV vs k-fold CV

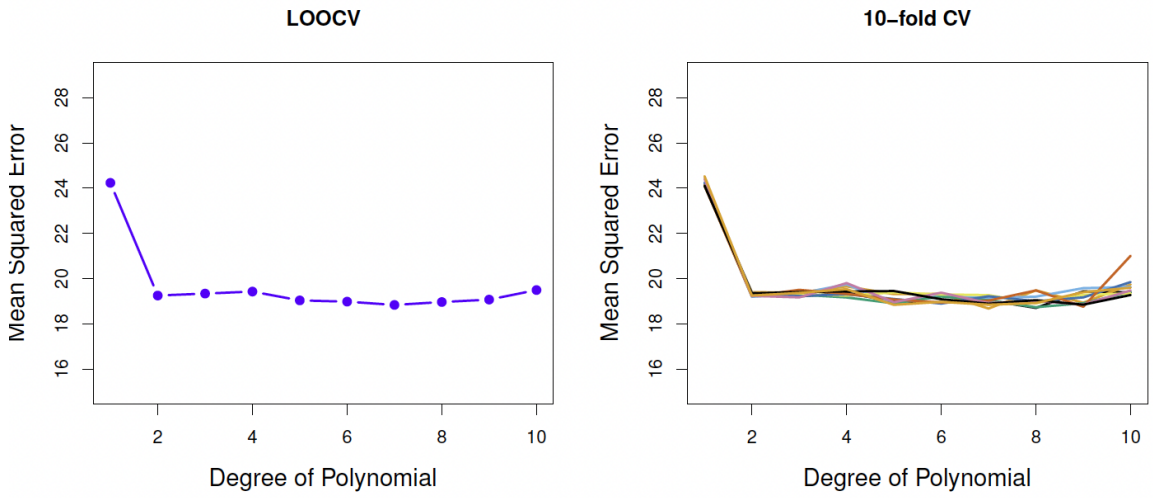


Figure: Mode selection with two CV methods[8].

[8] Gareth James, Daniela Witten, Trevor Hastie, Robert Tibshirani. "An introduction to statistical learning: with applications in R". Springer Texts in Statistics.

The Bootstap

A powerful tool that can quantify the uncertainty associated with a given learning method, e.g. it can estimate the standard error (SE) of $\hat{\beta}_k$, or MSE_{test}, ...

Main idea:
Given an original data-set $Z = D_n$, create $B > 1$ new datasets of size n:

- each new dataset Z^{*b} results from uniform sampling with replacement of the set Z.
- for each Z^{*b} calculate the unknown $\hat{\alpha}^{*b}$.

Resampled Data sets

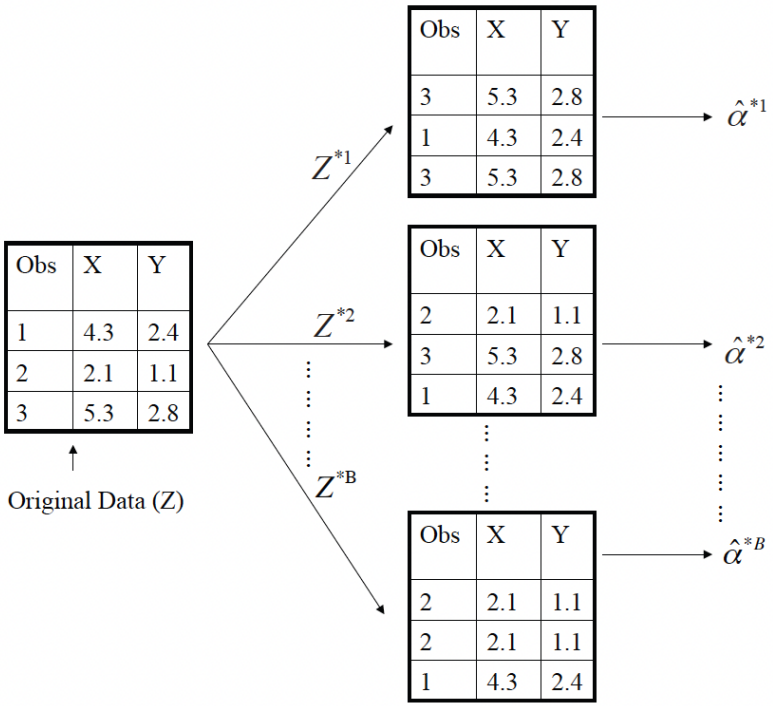


Figure: Example of Z^{*b} sets, $b = 1, 2, ..., B$ [9].

[9] Gareth James, Daniela Witten, Trevor Hastie, Robert Tibshirani. "An introduction to statistical learning: with applications in R". Springer Texts in Statistics.

Bootstrap estimates

- The Average

$$Av_B(\hat{\alpha}) = \bar{\alpha} = \frac{1}{B}\sum_{b=1}^{B}\hat{\alpha}^{*b}$$

- The Standard Error

$$SE_B(\hat{\alpha}) = \sqrt{\frac{1}{(B-1)}\sum_{b=1}^{B}(\hat{\alpha}^{*b} - \bar{\alpha})}$$

Resampled stats

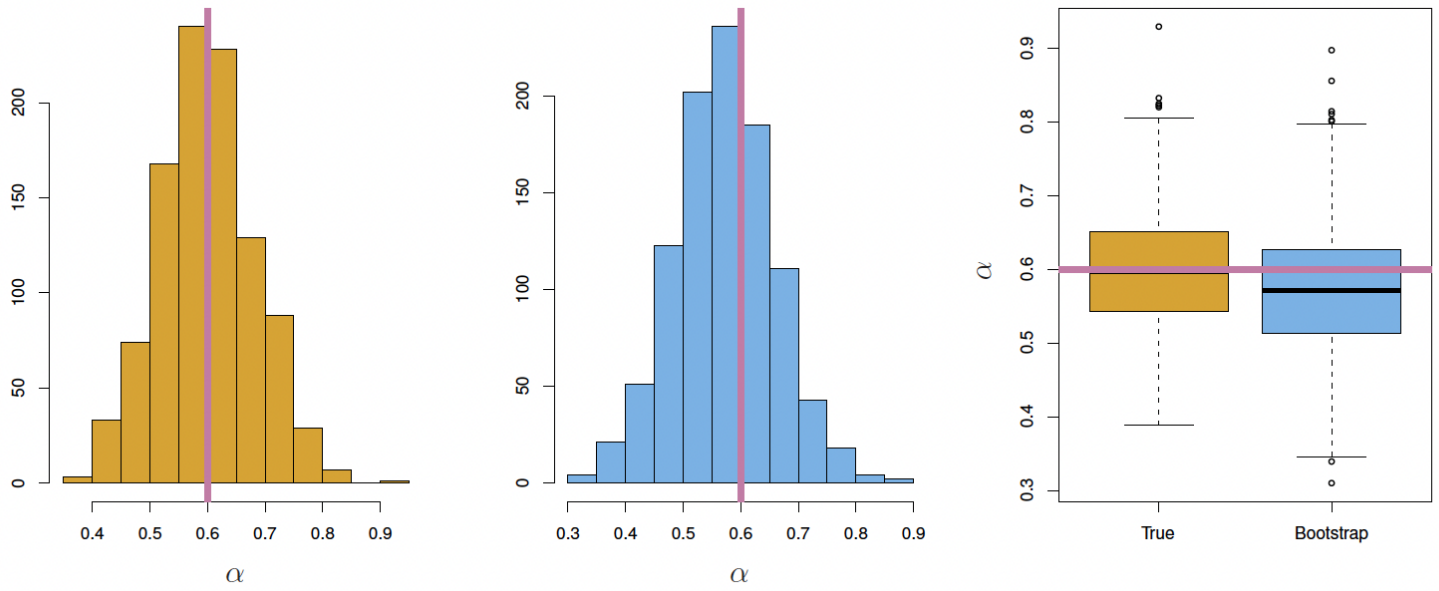


Figure: Real vs Bootstrapped statistics[10].

[10] Gareth James, Daniela Witten, Trevor Hastie, Robert Tibshirani. "An introduction to statistical learning: with applications in R". Springer Texts in Statistics.

Exercise

Consider the numerical example with $D_n = \{(1,3),(2,4),(3,8),(4,9)\}$.

Use this set and apply :

- Validation set approach
- LOOCV.
- 2-fold CV.
- Bootstrapping.